Languine

Supporting your Language Learning Journey

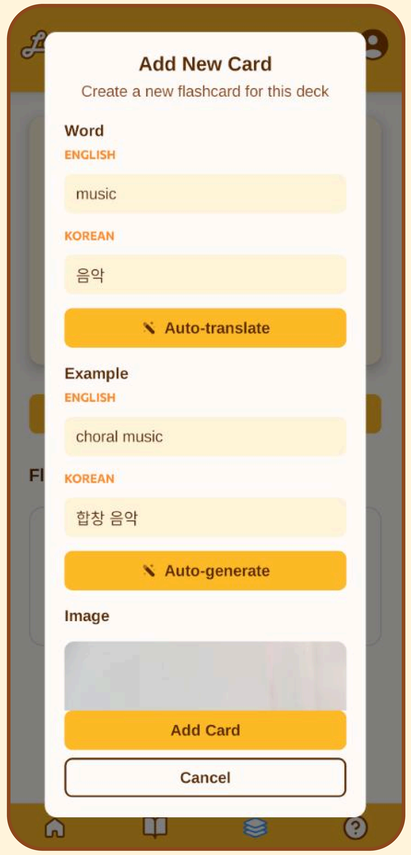
The Problem

Many language learners rely on flashcard apps to learn new vocabulary, but creating effective cards is often **tedious**. Looking up translations, pronunciations, images, and finding helpful examples takes **considerable effort**, causing learners to spend a large portion of their study time creating decks **instead of actually reviewing their flashcards**.

Our Solution

Languine is a mobile flashcard app supporting users in their vocabulary memorizing journey. It allows users to **quickly generate flashcards** for new words they encounter, and **review those flashcards** via a free spaced repetition scheduler (**FSRS**) algorithm that **auto-optimizes** after a number of reviews to better model each user's unique retention patterns.

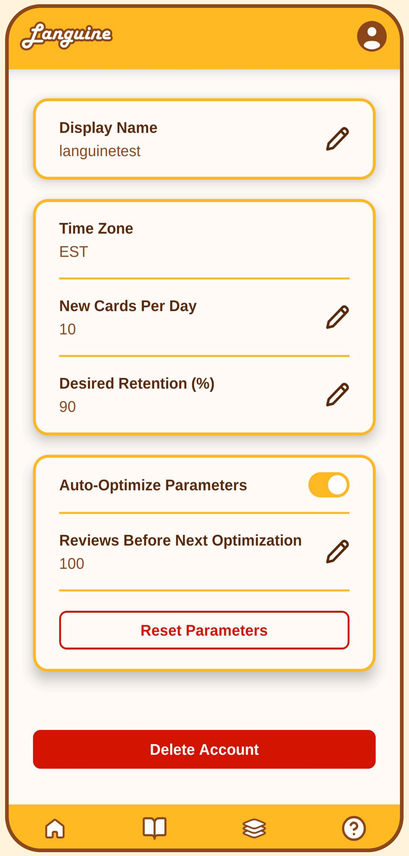
Add New Card



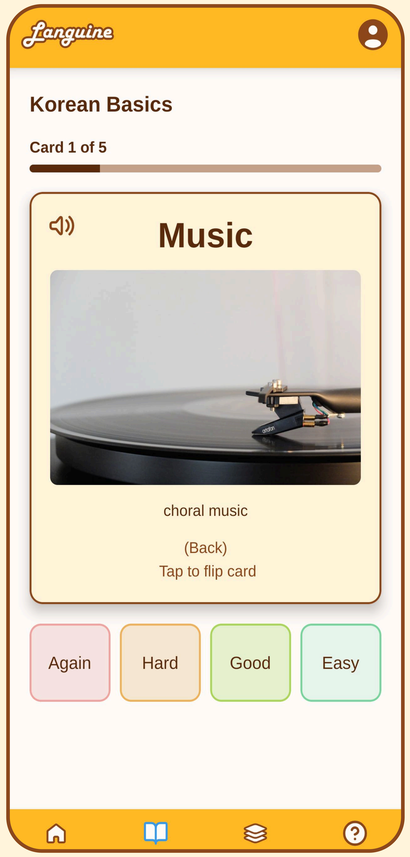
Card Generation

Users **enter a word** in their known language or target language, and the app will help **generate translations**, **example sentences**, and **images**. Once the card is created, **TTS is generated** in the background.

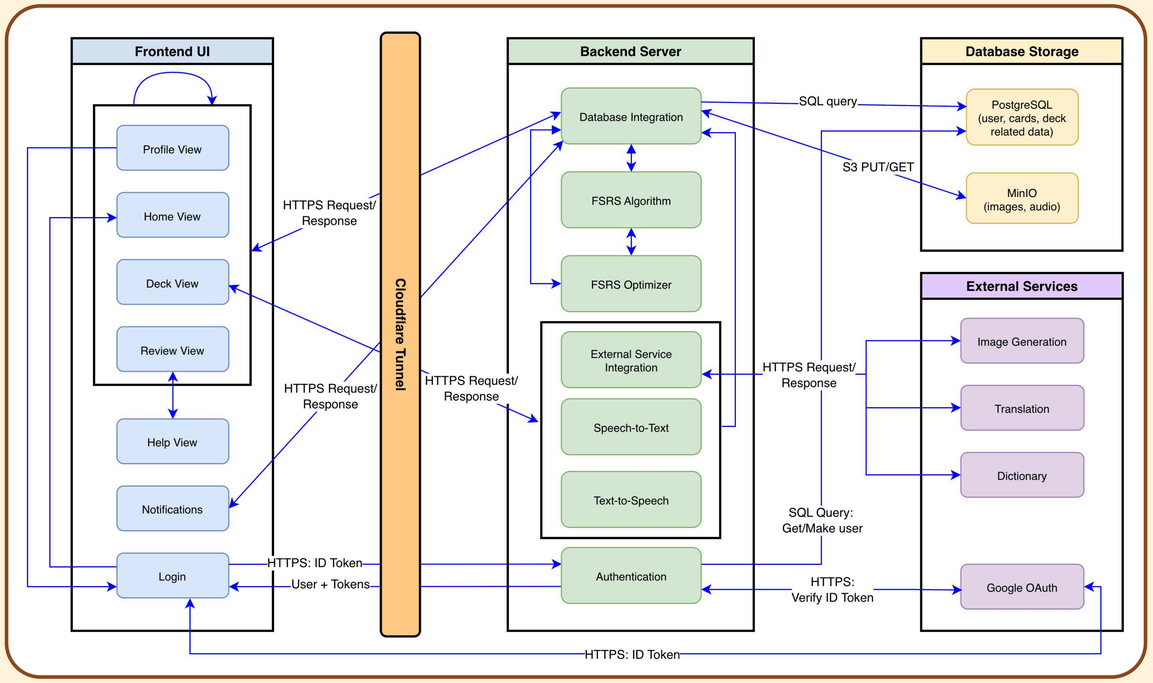
User Settings



Deck Revision



Components and Integration



Tech Stack

FSRS Algorithm & Optimizer

FSRS models **memory decay** by estimating a card's **stability**, **difficulty**, and **retrievability**. When a user reviews a card and assigns a **grade**, the algorithm updates these values to reflect how well the card is remembered. It then **predicts when the card should be shown next** to maintain the user's **desired retention**. Once the review threshold is met, the model **auto-optimizes its parameters** based on the user's **review history** to better match their personal memory patterns.

Our Team

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